



KNS INSTITUTE OF TECHNOLOGY, BENGALURU-64

ENGINEERING PHYSICS SOLVED NUMERICALS

AS PER 22 SCHEME

MODULE 1 - OSCILLATIONS AND WAVES (For MECH AND CIVIL STREAM ONLY)

SPRING CONSTANT:

1. A man weighing 600 N steps on a spring scale machine. The spring in the machine is compressed by 1 cm. Find the force constant of the spring.

Data: $W=600 \text{ N}$, $x=0.01 \text{ m}$.

To find $k = ?$

Solution:

$$k = -\frac{F_x}{x} = -\frac{W}{x} = \frac{-600}{-0.01} = 6 \times 10^4 \text{ N/m}$$

$\therefore$ Force constant of the spring, $k = 6 \times 10^4 \text{ N/m}$

2. A mass of 5 kg is suspended from the free end of a spring. When set for vertical oscillations, the system executes 40 oscillations in 100 seconds. Calculate the force constant of the spring.

Solution:

(Feb 2020- 18S)

$$\text{Period } T = \frac{100}{40} = 2.5 \text{ seconds}$$

We have $T = 2\pi \sqrt{\frac{m}{k}}$

Squaring & simplifying we have,

$$k = 4\pi^2 \frac{m}{T^2} = 4\pi^2 \times \frac{5}{2.5^2} = 31.5 \text{ N/m}$$

$\therefore$ The force constant of the spring, $k = 31.5 \text{ N/m}$

3. A mass 0.5 kg causes an extension 0.03m in a spring & the system is set for oscillations.

Find (i) Force constant k of the spring, (ii) Angular frequency, and

(iii) Period T of the resulting oscillation.

(Jan 2019 & July 2021 – 18S)

Data: $x = -0.03 \text{ m}$, $m = 0.5 \text{ kg}$

To find: k, ω , & T .

Solution: Force acting $F = mg = 0.5 \times 9.8 \text{ N}$

Restoring force, $F = -4.9 \text{ N}$

$$k = -\frac{F_x}{x} = \frac{-4.9}{-0.03} = 163.3 \text{ N/m}$$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{163.3}{0.5}}$$

$$= 18.1 \text{ rad/sec}$$

$$f = \frac{\omega}{2\pi} = \frac{18.1}{2\pi}$$

$$f = 2.877 \text{ Hz}$$

$$T = \frac{1}{f}$$

$$T = \frac{1}{2.877} = 0.35 \text{ sec}$$

$\therefore$ Force constant k of the spring, $k = 163.3 \text{ N/m}$

$\therefore$ Angular frequency, $\omega = 18.1 \frac{\text{rad}}{\text{sec}}$

$\therefore$ Period T of the resulting oscillation, $T = 0.35 \text{ sec}$

4. An electric motor weighing 50 kg is mounted on 4 springs each of which has a spring constant $2 \times 10^3 \text{ N/m}$. The motor moves only in vertical direction. Find the natural frequency of the system.

Data: $m = 50 \text{ kg}$ on 4 springs, $k = 2 \times 10^3 \text{ N/m}$

To find: $f = ?$

$$f = \frac{1}{2\pi} \sqrt{\frac{k_{\text{eff}}}{50}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{4 \times 2 \times 10^3}{50}} = 2 \text{ Hz}$$

$$f = 2 \text{ Hz}$$

$\therefore$ The natural frequency of the system, $f = 2 \text{ Hz}$

5. A spring undergoes an extension of 5cm for a load of 50gm. Find its frequency of oscillation, if it is set for vertical oscillations with a load of 200gm, attached to its bottom. Ignore the mass of the spring.

Data: Extension of the spring $x = 5 \times 10^{-2} \text{ m}$

Load acting during extension, $M = 50 \times 10^{-3} \text{ kg}$

Load acting during extension, $m = 200 \times 10^{-3} \text{ kg}$

To find: Frequency of oscillation of the spring, $\nu = ?$

Solution: Force constant $k = -\frac{F}{x} = \frac{50 \times 10^{-3} \times 9.8}{5 \times 10^{-2}} = 9.8 \text{ N/m}$

The wire executes oscillations with a frequency equal to its natural frequency ω in the absence of any external force.

$$\therefore \text{Angular frequency of the spring, } \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{9.8}{200 \times 10^{-3}}} = 7 \text{ rad/sec}$$

$$\therefore \text{Frequency of the oscillation of the spring, } v = \frac{\omega}{2\pi} = \frac{7}{2\pi} = 1.11 \text{ Hz}$$

$\therefore$ Frequency of the oscillation of the spring, $v = 1.11 \text{ Hz}$

6. A car has a spring system that supports the in-built mass 1000 kg. When a person with a weight 980 N sits at the C of G, the spring system sinks by 2.8 cm. When the car hits a bump, it starts oscillating vertically. Find the period and frequency of oscillation.

Data: $x = -0.028\text{m}$, inbuilt mass of car $m = 1000\text{kg}$

Person's weight $W = 980\text{N}$

To find: T and f When the person sits, the weight acting is 980 N.

We have, $F_x = -kx$ or $k = -\frac{F_x}{x} = \frac{-980}{-0.028} = 3.5 \times 10^4 \text{ N/m}$

$$\text{Person's mass} = \frac{F}{g} = \frac{980}{9.8} = 100\text{kg}$$

Total mass = 1000 + 100 = 1100 kg.

$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{1100}{3.5 \times 10^4}}$$

$$T = 1.11 \text{ seconds}$$

$$f = \frac{1}{T} = \frac{1}{1.11} = 0.9\text{Hz}$$

$\therefore$ The Period and frequency of oscillation, $f = 0.9\text{Hz}$

7. A car has a spring system that supports the in-built mass 1000 kg. When a person with a weight 980 N sits at the C of G, the spring system sinks by 2.8 cm. When the car hits a bump, it starts oscillating vertically. Find the period and frequency of oscillation.

Data: $x = -0.028\text{m}$, inbuilt mass of car $m = 1000\text{kg}$

Person's weight $W = 980\text{N}$

To find: T and f when the person sits, the weight acting is 980 N.

We have, $F_x = -kx$ or $k = -\frac{F_x}{x} = \frac{-980}{-0.028} = 3.5 \times 10^4 \text{ N/m}$

$$\text{Person's mass} = \frac{F}{g} = \frac{980}{9.8} = 100\text{kg}$$

Total mass = 1000 + 100 = 1100 kg.

$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{1100}{3.5 \times 10^4}}$$

$$T = 1.11 \text{ seconds}$$

$$f = \frac{1}{T} = \frac{1}{1.11} = 0.9\text{Hz}$$

$\therefore$ The Period and frequency of oscillation, $f = 0.9\text{Hz}$

8. Calculate the resonance frequency for a simple pendulum of length 1m. (Jan 2020 – 18S)

Data: Length of the simple pendulum, $L = 1\text{m}$

To find: Resonance frequency of oscillation, $\nu = ?$

Solution:

A simple pendulum set for oscillation, oscillates with a period.

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$T = 2\pi \sqrt{\frac{1}{9.8}}, \quad T = 2 \text{ seconds}$$

$$\therefore \text{Frequency of the oscillation, } \nu = \frac{1}{T} = \frac{1}{2} = 0.5 \text{ Hz}$$

9. Evaluate the resonance frequency of a spring of force constant 1974 N/m, carrying of mass of 2kg. (Jan 2021 & Feb 2022- 18S)

Data: Force constant, $k = 1974 \text{ N/m}$

Mass suspended $m = 2\text{kg}$

To find:

Resonance frequency $\nu = ?$

Solution: We know that resonance frequency of any system is equal to its natural frequency of oscillation.

$$\text{But the natural frequency, } \omega = \sqrt{\frac{k}{m}} \text{ and also } \omega = 2\pi\nu$$

$$\text{Resonance frequency, } \nu = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{1974}{2}} = 5\text{Hz}$$

$$\therefore \text{Resonance frequency of the spring, } \nu = 5\text{Hz}$$

10. An automobile of mass 2000 kg is supported by 4 equal shock absorbing springs. It executes underdamped oscillations (which is a bad design of course) of period 0.8second when bumped. Evaluate the force constant of the shock absorber Also, evaluate the period of oscillation when four persons each of weight 60 kg board the automobile.

Data: Mass of the automobile, $M = 2000 \text{ kg}$.

No. of supporting shock absorbers, $n = 4$.

Period of oscillation, $T_1 = 0.8 \text{ second}$.

Mass of each person getting into the automobile, $M' = 60 \text{ kg}$

No. of persons, $N=4$.

To find: The force constant of the absorber, $k = ?$

The period of oscillation after 4 people get into the automobile, $T_2 = ?$

Solution:

$$\text{Mass supported by each of the springs, } m = \frac{M}{n} = \frac{2000}{4} = 500 \text{ kg}$$

$$\text{We have, equation for natural frequency of oscillation, } \omega = \sqrt{\frac{k}{m}}$$

But $\omega = \frac{2\pi}{T_1}$

$$T_1 = 2\pi \sqrt{\frac{m}{k}}$$

$$0.8 = 2\pi \sqrt{\frac{500}{k}}$$

$$k = \left[\frac{4\pi^2}{0.8^2} \right] \times 500 = 30842$$

Since 4 persons each of mass 60 kg are added to the automobile, extra mass supported by each shock absorber 60 kg.

Total mass supported by each shock absorber = 560 kg.

New period of oscillation T_2 is,

$$T_2 = 2\pi \sqrt{\frac{560}{30842}} = 0.847 \text{ second}$$

$$\therefore \text{Force constant of the shock absorber} = 30842 \frac{N}{m}$$

And the Period of oscillation after the people get into the car = 0.847 second

OSCILLATIONS:

11. A free particle is executing simple harmonic motion in a straight line. The maximum velocity it attains during any oscillation is 62.8m/s. find the frequency of the oscillation, if its amplitude is 0.5m. (July 2021-18S)

Data: Maximum velocity during oscillation, $v_{max} = 62.8 \text{ m/s}$

Amplitude, $a = 0.5\text{m}$

To find: Frequency of the oscillation, $\nu = ?$

Solution: *We have the equation for free vibration, $x = a \sin \omega t$*

velocity is given by, $v = \frac{dx}{dt} = a\omega \cos \omega t$

$$\text{or } v = a\omega \sqrt{1 - \sin^2 \omega t}$$

$$\text{or } v = \omega \sqrt{a^2 - a^2 \sin^2 \omega t} \quad \text{i.e. } v = \omega \sqrt{a^2 - x^2}$$

The particle attains maximum velocity while passing through its equilibrium position, at which the time and displacement is zero (i.e., $x = 0$)

$$\therefore v_{max} = \omega \sqrt{a^2 - 0} = \omega a$$

$$\therefore \text{angular frequency, } \omega = \frac{v_{max}}{a} = \frac{62.8}{0.5} = 125.6 \frac{\text{rad}}{\text{sec}}$$

$$\therefore \text{Frequency of oscillation, } \nu = \frac{\omega}{2\pi} = \frac{125.6}{2\pi} = 20\text{Hz}$$

The frequency of oscillation of the particle is 20Hz.

12. A free particle is executing simple harmonic motion in a straight line with a period of 25 seconds, 5 seconds after it has crossed the equilibrium point, the velocity is found to be 0.7m/s. find its displacement at the end of 10 seconds and also the amplitude of oscillation. **(Jan 2019 & July 2022 – 18S)**

Data: Period of oscillation, $T = 25$ sec,
Velocity, $v_1 = 0.7\text{m/s}$ at time $t_1 = 5\text{sec}$ after crossing equilibrium position.

To Find: Amplitude of oscillation, $a = ?$
Displacement x , at time $t_2 = 10$ sec

Solution: Angular frequency, $\omega = \frac{2\pi}{T} = \frac{2\pi}{25}$

We have the equation for displacement, $x = a \sin \omega t$

$$\therefore \text{Velocity}, v = \frac{dx}{dt} = a\omega \cos \omega t$$

At time $t_1 = 5$ seconds, $v = 0.7$ m/s

$$\therefore \text{Velocity}, v = a\omega \cos \omega t$$

$$\text{substitute the values, } 0.7 = a \times \frac{2\pi}{25} \times \cos \left(\frac{2\pi}{25} \times 5 \right)$$

$$\text{Note: [in radian mode], } \cos \left(\frac{2\pi}{5} \right) = \cos(1.257) = 0.309$$

$$[\text{in degree mode}], \cos \left(\frac{2 \times 180}{5} \right) = 0.309$$

$$0.7 = 0.25 \times a \times 0.309$$

$$\therefore a = 9.06\text{m}$$

$\therefore$ Displacement at the end of 10 seconds is obtained by the formula, $x = a \sin \omega t$

$$x = 9.06 \times \sin \left(\frac{2\pi}{25} \times 10 \right) = 9.06 \times 0.588 = 5.3\text{m}$$

$\therefore$ The displacement at the end of 10 seconds is 5.3m, and the amplitude is 9.06m.

13. A 20gm oscillator with natural angular frequency 10 rad/second is vibrating in damping medium. The damping force is proportional to the velocity of the vibrator. If the damping coefficient is 0.17, how does the oscillator decay? **(July 2019, Aug 2020, July 22 – 18S)**

Data: Mass of the oscillator, $m = 20\text{gm} = 20 \times 10^{-3}\text{kg}$
Natural angular frequency, $\omega = 10 \frac{\text{rad}}{\text{second}}$

Damping force $\propto \left(\frac{dx}{dt} \right)$, i.e.,

Damping force $= r \left(\frac{dx}{dt} \right)$, where r is the damping coefficient,

Damping coefficient, $r = 0.17 \text{ kg/s}$

To find: To identify the case under which the decay of oscillations takes place.

Solution: The equation of motion for damped oscillation is,

$$m \frac{d^2x}{dt^2} + r \frac{dx}{dt} + kx = 0$$

$$\text{or } \frac{d^2x}{dt^2} + \frac{r}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

$$\text{Let } \frac{r}{m} = 2b \dots \dots \dots (1) \quad \text{and} \quad \frac{k}{m} = \omega^2 \dots \dots \dots (2)$$

Where ω is the natural angular frequency.

We know that, the nature of damping is decided by comparing the values of b^2 and ω^2 .

$$\text{Now, } b = \frac{r}{2m} \quad \text{from eqn (1)}$$

$$\therefore b = \frac{0.17}{2 \times 0.02} = 4.25$$

$$b^2 = 4.25^2 = 18.06$$

$$\text{from the above data, } \omega = 10 \frac{\text{rad}}{\text{sec}} \quad \text{or}$$

$$\omega^2 = 100$$

comparing the values of b^2 and ω^2 , we find that $b^2 < \omega^2$

$\therefore$ It is the case of underdamping.

14. A mass of 2 kg suspended by a spring of force constant 51.26 N/m is executing damped simple harmonic oscillations with damping of 5 kg/s. Identify whether it is the case of underdamping or of over damping. Also estimate the value of damping required for the oscillations to be critically damped (Ignore the mass of the spring). (July - 2022- 21S)

Data: Mass suspended $m = 2$ kg.
Force constant of the spring, $k = 51.26$ N/m.
Damping factor, $r = 5$ kg/s

To find:

- To identify whether it is the case of underdamping or of over damping.
- Value of r , required for critical damping?

Solution: The natural frequency of oscillations is given by, $\omega = \sqrt{\frac{k}{m}}$

$$\omega^2 = \frac{k}{m} = \frac{51.26}{2} = 25.63 \dots \dots \dots (1)$$

$$\text{Now, } b = \frac{r}{2m}$$

$$b^2 = \frac{r^2}{4m^2} = \frac{5^2}{4 \times 2^2} = 1.5625 \dots \dots \dots (2)$$

Comparing equation (1) and (2), we notice that, $b^2 < \omega^2$

Hence it is the case of underdamping.

For critical damping, $b^2 = \omega^2$

$$\frac{r^2}{4m^2} = \omega^2$$

$$\text{or } r = 2m\sqrt{\omega^2} = 2 \times 2\sqrt{25.63}$$

$$\therefore r = 20.25 \text{ Kg/s}$$

$\therefore$ The value of damping required for the oscillation to be critically damped is 20.25 kg/s

15. Calculate the peak amplitude of vibration of a system whose natural frequency is 1000Hz when it oscillates in a resistive medium for which the value of damping/unit mass is 0.008rad/s under the action of external periodic force /unit mass of amplitude 5N/Kg, with tunable frequency.

Data: Natural frequency, $\nu = 1000\text{Hz}$,

(Feb 2022 -21S)

Amplitude of the force unit mass, $F/m = 5 \text{ N/kg}$,

Damping coefficient, $r/m = 0.008 \text{ rad/s}$.

To find: Maximum amplitude of vibration, $a_{\max} = ?$

Solution: $b = \frac{r}{2m} = \frac{1}{2} \times 0.008 = 0.004$

We have the equation for amplitude, $a = \frac{\frac{F}{m}}{\sqrt{((\omega^2 - p^2)^2 + 4b^2 p^2)}}$

Maximum amplitude is achieved when the frequency p of the applied force is tuned to match the natural frequency ω of the system.

i.e., $p = \omega$ or $p^2 = \omega^2$

$$a_{\max} = \frac{\frac{F}{m}}{\sqrt{0+4b^2 p^2}} = \frac{\frac{F}{m}}{2bp}$$

$$\text{since w.k.t } \frac{F}{m} = \frac{5\text{N}}{\text{kg}}$$

$$\text{Now } p = 2\pi\nu = 2\pi \times 1000$$

and

$$b = 0.004$$

$$a_{\max} = \frac{5}{2 \times 0.004 \times 2\pi \times 1000} = \frac{5}{50} = 0.1\text{m}$$

$\therefore$ The maximum amplitude of vibration = 0.1m

16. A vibrating system of natural frequency 500 Cycles/ second, is forced to vibrate with a periodic force/unit mass of amplitude 100×10^{-5} Newton/kg in the presence of a damping/ unit mass of 0.01×10^{-3} rad/s. Calculate the maximum amplitude of vibration of the system. (Jan 2021 - 18S)

Data:

Natural frequency, $\nu = 500 \text{ cycles/second}$,

Amplitude of the force unit mass, $F/m = 100 \times 10^{-5} \text{ N/kg}$,

Damping coefficient, $r/m = 0.01 \times 10^{-3} \text{ rad/s}$.

To find: Maximum amplitude of vibration, $a_{\max} = ?$

Solution:

The system attains maximum amplitude when it vibrates as its natural frequency for which the equation is,

$$a_{\max} = \frac{F/m}{2b\omega}$$

$$\text{Where } b = \frac{r}{2m} = \frac{0.01 \times 10^{-3}}{2} = 0.005 \times 10^{-3}$$

And, the natural (angular) frequency, ω is $\omega = 2\pi\nu = 2\pi \times 500$

$$a_{\max} = \frac{100 \times 10^{-5}}{2 \times 0.005 \times 10^{-3} \times 2\pi \times 500}$$

$$a_{\max} = 0.318\text{m}$$

$\therefore$ The maximum amplitude of vibration of the system is 0.318 meter.

SHOCK WAVES:

1. The distance between the two pressure sensors in a shock tube is 150 mm. The time taken by a shock wave to travel this distance is 0.3 ms. If the velocity of sound under the same condition is 340 ms^{-1} . Find the Mach number of the shock.

Data:

Distance between the two pressure sensors in a shock tube, $d = 150 \text{ mm}$ i.e. $d = 150 \times 10^{-3} \text{ m}$

Time taken by a shock wave to travel this distance $t = 0.3 \text{ milliseconds}$ or $t = 0.3 \times 10^{-3} \text{ sec}$

Velocity of sound, $a = 340 \text{ ms}^{-1}$

$$\text{Speed of shock waves } v = \frac{\text{distance}}{\text{time}} = \frac{150 \times 10^{-3}}{0.3 \times 10^{-3}} = 500 \text{ ms}^{-1}$$

$$\text{Mach number } M = \frac{v}{a} = \frac{500}{340} = 1.47$$

$\therefore$ Mach number of the shock waves is given by 1.47.

2. The distance between two pressure sensors in a shock tube is 100 mm. The time taken by a shock wave to travel this distance is 200 μs . If the velocity of sound under the same conditions is 340 m/s, find the Mach number of the shock wave. Ans: M = 1.47
3. In a Reddy tube experiment, it was found that, the time taken to travel between the two sensors is 195 μs . If the distance between the two sensors is 100 mm, find the Mach number. Ans: M=1.51.
4. The distance between the two pressure sensors in shock tube 100 mm. The time taken by a shock wave to travel this distance is 100 microsecond. If the velocity of sound under the same conditions is 340 ms^{-1} . Find the Mach number of the shock wave. Ans: M 1.51

MODULE 2 –QUANTUM MECHANICS (FOR CSE AND ECE STREAM ONLY)

DE-BROGLIE WAVELENGTH:

1. Calculate the de-Broglie wavelength associated with neutron of mass 1.674×10^{-27} kg moving with one tenth part of the velocity of light. (DEC 2018 – 17S)

Data: Mass of the neutron: 1.674×10^{-27} kg,

Velocity of light, $c = 3 \times 10^8$ m/s

Velocity of neutron $v = \frac{1}{10} \times c = 3 \times 10^7$ m/s

To Find: De Broglie wavelength, $\lambda = ?$

Solution: De Broglie wavelength, $\lambda = \frac{h}{mv}$

$$\lambda = \frac{6.6 \times 10^{-34}}{(1.074 \times 10^{-27})(3 \times 10^7)} = 1.314 \times 10^{-14} \text{ m}$$

$\therefore$ De – Broglie wavelength associated of the neutron is 1.314×10^{-14} m

2. A particle of mass $0.5 \text{ MeV}/c^2$ has kinetic energy 100eV. Find its de-Broglie wavelength.

Data: Mass of the particle $m = 0.5 \text{ MeV}/c^2$,

[Dec. 2014 | Jan 2018, Dec 2019 | Feb 22 – 18S].

Kinetic energy $E = 100 \times 1.609 \times 10^{-19}$ J

Velocity of light, $c = 3 \times 10^8$ m/s

Velocity of neutron $v = \frac{1}{10} \times c = 3 \times 10^7$ m/s

To find: De Broglie wavelength, $\lambda = ?$

Solution: De Broglie wavelength, $\lambda = \frac{h}{mv}$

We have, mass of the particle, $m = 0.5 \frac{\text{MeV}}{c^2} = \frac{0.5 \times 10^6 \times 1.602 \times 10^{-19}}{3 \times 10^8} = 8.9 \times 10^{-31} \text{ kg}$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 8.9 \times 10^{-31} \times 100 \times 1.602 \times 10^{-19}}} = 1.24 \times 10^{-10} \text{ m}$$

$\therefore$ The De Broglie wavelength of the particle is 1.24×10^{-10} m

3. Evaluate De Broglie Wavelength of helium nucleus that is accelerated through 500V. (Feb.22 – 21S)

Data: Mass of proton = mass of neutron = 1.67×10^{-27} kg.

Potential difference = 500V

To Find: De Broglie wavelength, $\lambda = ?$

Solution: A Helium nucleus has 2 protons and 2 neutrons.

$\therefore$ Mass of nuclues = $4m \times 1.67 \times 10^{-27} \text{ kg}$

since there are 2 protons, the charge on the nuclues, $q = 2 \times 1.602 \times 10^{-19}$

we have the equation for de broglie wavelength, $\lambda = \frac{h}{\sqrt{2mE}}$

But, for a charge of q accelerated through a potential V , the energy is given by $E = qV$ (assuming its initial velocity as zero).

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2(4 \times 1.67 \times 10^{-27})(2 \times 1.602 \times 10^{-19}) \times 500}} = 4.53 \times 10^{-13} \text{ m}$$

$\therefore$ The de broglie wavelength is 4.53×10^{-13} m

HEISENBERG'S UNCERTAINTY PRINCIPLE:

4. The inherent uncertainty in the measurement of time spent by Iridium 191 nuclei in the excited state is found to be 1.4×10^{-10} sec. estimate the uncertainty that result in its energy in the excited state.

Data: uncertainty in time $\Delta t = 1.4 \times 10^{-10}$ sec

(Sep 20 | Feb 21– 18S)

To find: Uncertainty in energy $\Delta E = ?$

By Heisenberg's uncertainty principle, $\Delta E = \frac{h}{4\pi\Delta t}$

$$\Delta E = \frac{6.63 \times 10^{-34}}{4\pi \times 1.4 \times 10^{-10}} = 3.77 \times 10^{-25} \text{ J}$$

$$\Delta E = 2.35 \times 10^{-6} \text{ eV}$$

$$\therefore \text{Uncertainty in energy } \Delta E = 2.35 \times 10^{-6} \text{ eV}$$

5. A spectral line of Wavelength $5461 \text{ \AA}$ has a width of $10^{-4} \text{ \AA}$. Evaluate the minimum time spent by the electron in the upper energy state.

(Aug 22 – 18S)

Data: wavelength of the spectral line $\lambda = 5461 \times 10^{-10} \text{ m}$

Width of the spectral lines (also means uncertainty in the wavelength value),

$$\Delta\lambda = 10^{-4} \times 10^{-10} \text{ m} = 10^{-14} \text{ m}$$

To find:

The minimum time sent by the electrons in the upper state (also means the uncertainty in the measurement of time for the system), $\Delta t = ?$

Solution : We have the equation $E = hv = \frac{hc}{\lambda}$

$$\Delta E = hc \Delta \left[\frac{1}{\lambda} \right] \text{ or}$$

$$\Delta E = hc \frac{\Delta\lambda}{\lambda^2} \dots \dots \dots (1)$$

As per the uncertainty principle. $\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$

$$\text{Or } \Delta t \geq \frac{h}{4\pi \cdot \Delta E} \dots \dots \dots (2)$$

Sing eqn(1), the right side of the above inequality can be written as,

$$\begin{aligned} \Delta t &= \frac{h}{4\pi \cdot \Delta E} = \frac{h\lambda^2}{4\pi(hc \cdot \Delta\lambda)} = \frac{\lambda^2}{4\pi(c \cdot \Delta\lambda)} \\ &= \frac{(5461 \times 10^{-10})^2}{4\pi \times 3 \times 10^8 \times 10^{-14}} \\ \Delta t &= 0.8 \times 10^{-8} \text{ second.} \end{aligned}$$

Thus, Using the above in the inequality (2), we have,

$$\Delta t = 0.8 \times 10^{-8} \text{ second}$$

$\therefore$ The minimum time spent by the electrons is, $\Delta t = 0.8 \times 10^{-8} \text{ second}$

6. The position and momentum of an electron with energy 0.5 KeV are determined. What is the minimum percentage of uncertainty in its momentum, if the uncertainty in the measurement of its position is $0.5 \text{ \AA}$.

Data:

(Feb.21-21S)

Energy of electron, $E = 0.5 \times 10^3 \text{ eV} = 0.5 \times 10^3 \times 1.602 \times 10^{-19} \text{ J}$

Maximum uncertainty in position, $\Delta x = 0.5 \text{ \AA} = 0.5 \times 10^{-10} \text{ m}$

To find:

Minimum percentage uncertainty in momentum =?

Solution: We have the uncertainty principle given by, $\Delta x \cdot \Delta P_x \geq \frac{h}{4\pi}$

$$\Delta P_x \geq \frac{h}{4\pi \times \Delta x} = \frac{6.63 \times 10^{-34}}{4\pi \times 0.5 \times 10^{-10}} = 1.06 \times 10^{-24}$$

$\therefore$ Uncertainty in momentum = 1.06×10^{-24}

We have the equation for momentum, $P = \sqrt{2mE} = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.6 \times 10^{-19}}$

$\therefore$ Momentum $P = 1.207 \times 10^{-23}$

$\therefore$ Percentage uncertainty in momentum = $\frac{\text{Uncertainty in momentum}}{\text{momentum}} \times 100$

$$= \frac{1.06 \times 10^{-24}}{1.207 \times 10^{-23}} \times 100 = 8.8$$

$\therefore$ **Percentage of uncertainty = 8.8**

APPLICATION OF SCHROEDINGER WAVE EQUATION:

7. An electron bound in a one dimensional potential well of width $1 \text{ \AA}$, but infinite well height. Find its energy values in the ground state and in the first two excited states. (Jan.2019 | July2019 | July 2021- 18S)

Data: Width of the potential well, $a = 1 \text{ \AA} = 10^{-10} \text{ m}$

To find: Values of energy of the electron in the ground and the first two excited states

i.e $E_1 = ?$, $E_2 = ?$, $E_3 = ?$

Solution: The energy of the electron in a one dimensional infinite potential well is given by,

$$E_n = \frac{n^2 h^2}{8ma^2}$$

Where, the values of E that we get for $n=1, 2$ and 3 correspond respectively to E_1, E_2 and E_3 .

$$\therefore E_1 = \frac{n^2 h^2}{8ma^2} = \frac{1^2 (6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(10^{-10})^2}$$

$$\therefore E_0 = E_1 = 6.0314 \times 10^{-18} \text{ J} = 37.64 \text{ eV}$$

$$E_2 = \frac{n^2 h^2}{8ma^2} = \frac{2^2 (6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(10^{-10})^2} = 4 \times 6.0314 \times 10^{-18} \text{ J}$$

$$\therefore E_2 = 2.4125 \times 10^{-17} \text{ J} = 150.54 \text{ eV}$$

$$E_3 = \frac{n^2 h^2}{8ma^2} = \frac{3^2 (6.63 \times 10^{-34})^2}{8(9.11 \times 10^{-31})(10^{-10})^2} = 9 \times 6.0314 \times 10^{-18} \text{ J}$$

$$\therefore E_3 = 5.4282 \times 10^{-17} \text{ J} = 338.7 \text{ eV}$$

$\therefore$ The values of Energy of the electron in the ground and the first two excited states are *respectively to*

$$E_1 = 6.0314 \times 10^{-18} \text{ J or } 37.64 \text{ eV ,}$$

$$E_2 = 2.4125 \times 10^{-17} \text{ or } 150.54 \text{ eV and}$$

$$E_3 = 5.4282 \times 10^{-17} \text{ J or } 338.7 \text{ eV}$$

8. An electron bound in a one dimensional potential well of height and width of **0.2nm**. Calculate energy required for the ground state and its first two excited states. (Aug.2020 – 18S, July.2022 -21S)

NOTE: SOLVE AS PER PROBLEM NO.07

MODULE 3 – LASERS AND OPTICAL FIBRE-(FOR CIVIL, CSE AND ECE STREAM)

LASER:

1. A pulsed laser emits photon of wavelength 780nm with 20mW average power/pulse. Calculate the number of photons contained in each pulse if the pulseduration is 10 ns. [Jan 2019]

Data: Wavelength of photon, $\lambda = 780\text{nm} = 780 \times 10^{-9}\text{m}$

Power of each pulse $P = 20\text{mW} = 20 \times 10^{-3}\text{W}$

Duration of Each Pulse, $t = 10\text{ns} = 10 \times 10^{-9}\text{sec}$

To find: No. of photons in each pulse, $N=?$

Solution: wavelength of photon, $\lambda = 780\text{nm} = 780 \times 10^{-9}\text{m}$

$$\therefore \text{Energy of each photons, } \Delta E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{780 \times 10^{-9}} = 2.55 \times 10^{-19}\text{J}$$

Now, we have the energy of each pulse $E = \text{Power} \times \text{duration of pulse}$

$$E = P \times t = 20 \times 10^{-3} \times 10 \times 10^{-9} = 2 \times 10^{-10}\text{J}$$

If N is the number of photons (each of energy ΔE) in the pulse, then $N \times \Delta E = E$

$$\therefore N = \frac{E}{\Delta E} = \frac{2 \times 10^{-10}}{2.55 \times 10^{-19}} = 7.86 \times 10^8$$

$\therefore$ Number of photons in each pulse is 7.86×10^8 .

2. The average output of laser source emitting a laser beam of wavelength 632.8nm is 5Mw. Find the number of photons emitted per second by the laser source. (July 2019 | Aug 2021 | Feb 2022 – 18S)

Data : Wavelength of laser, $\lambda = 632.8\text{nm} = 632.8 \times 10^{-9}\text{m}$

Power of each pulse $P = 5\text{mW} = 5 \times 10^{-3}\text{W}$

To find : No. of photons in each pulse, $N=?$

Solution: we have $N = \frac{E}{\Delta E}$ ($\therefore E = P \times t$, $\Delta E = \frac{hc}{\lambda}$)

$$N = \frac{P \times t}{\frac{hc}{\lambda}} \text{ or } N = \frac{P \times t \times \lambda}{hc} = \frac{5 \times 10^{-3} \times 632.8 \times 10^{-9}}{6.63 \times 10^{-34} \times 3 \times 10^8} = 1.59 \times 10^{16}$$

$\therefore$ Number of photons in each pulse is 1.59×10^{16}

3. The ratio of population of two energy levels is 1.059×10^{-30} . Find the wavelength of light emitted by spontaneous emission at 330K. (Jan.2021 | July.2022 -18S)

Data: The ratio of population, $(N_2/N_1) = 1.059 \times 10^{-30}$

Ambient temperature, $T = 330\text{K}$

To find : Wavelength of light emitted,, $\lambda = ?$

Solution: we have Boltzmann factor, $\frac{N_2}{N_1} = e^{\frac{-\Delta E}{kT}}$

By Taking natural log on both sides, we have

$$\ln \left(\frac{N_2}{N_1} \right) = \frac{-\Delta E}{kT} = \frac{-hc}{\lambda kT} = - \left(\frac{hc}{k} \right) \left(\frac{1}{\lambda T} \right)$$

$$\text{since } \left(\frac{hc}{k} \right) = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.38 \times 10^{-23}} = 0.014413$$

$$\lambda = - \left(\frac{hc}{k} \right) \left(\frac{1}{\ln \left(\frac{N_2}{N_1} \right) T} \right) = \frac{0.014413}{\ln(1.059 \times 10^{-30}) \times 330} = 632 \times 10^{-9} \text{m} = 632 \text{nm}$$

$\therefore$ The wavelength of light emitted by spontaneous emission is 632nm.

4. A pulse from laser with power 1mW lasts for 10ns, if the number of photons emitted per pulse is 3.491×10^7 . Calculate the wavelength of the laser. **(Feb 2022 – 18S)**

Data: Power per pulse $P = 1 \text{mW} = 1 \times 10^{-3} \text{W}$

No. of photons in each pulse, $N = 3.491 \times 10^7$

Duration of Each Pulse, $t = 10 \text{ns} = 10 \times 10^{-9} \text{sec}$

To find: wavelength of laser, $\lambda = ?$

Solution: $N = \frac{E}{\Delta E}$ ($\therefore E = P \times t$, $\Delta E = \frac{hc}{\lambda}$)

$$N = \frac{P \times t}{\frac{hc}{\lambda}} \text{ or } N = \frac{P \times t \times \lambda}{hc}$$

$$\text{Rearranging this, we get } \lambda = \frac{N \times h \times c}{P \times t} = \frac{3.491 \times 10^7 \times 6.63 \times 10^{-34} \times 3 \times 10^8}{1 \times 10^{-3} \times 10 \times 10^{-9}} = 6943 \times 10^{-10} \text{m} = 6943 \text{\AA}$$

$\therefore$ wavelength of laser, $\lambda = 6943 \text{\AA}$

5. How many photons of yellow light of wavelength $5500 \times 10^{-10} \text{m}$ constitutes 1.5J of energy.

Data: Wavelength of yellow light, $\lambda = 5500 \times 10^{-10} \text{m}$

(July 2022 – 18S)

Energy, $E = 1.5 \text{J}$

To find : No of photons $n = ?$

$$\text{Solution: } E = \frac{N \times hc}{\lambda} \text{ or by rearranging, } N = \frac{E \times \lambda}{hc} = \frac{1.5 \times 5500 \times 10^{-10}}{6.63 \times 10^{-34} \times 3 \times 10^8} = 4.1478 \times 10^{18}.$$

No. of photons emitted N is 4.1478×10^{18} .

6. A He-Ne laser is emitting a beam with an average power of 4.5mW. Find the number of photons emitted per second by the laser. The wavelength of the light is $6328 \text{\AA}$.

(Refer problem no.2 same type)

(Answer $N = 1.43 \times 10^{16}$)

7. A Ruby laser a pulse of 20ns duration with average power per pulse being 100KW. If the number of photons in each pulse is 6.981×10^{15} . Calculate the wavelength of photons.

(Refer problem No.4 same type)

(Answer $\lambda = 6943 \times 10^{-10} \text{m}$)

8. Find the ratio of population of two energy levels in a laser if the transition between them produced a wavelength of 694.3nm. Assume the ambient temperature is 27°C .

Data: wavelength of light emitted, $\lambda = 694.3 \text{nm}$ or $\lambda = 694.3 \times 10^{-9} \text{m}$

Ambient temperature, $T = 27^\circ \text{C}$

or $(27+273) = 300 \text{K}$

To find: The ratio of population, $(N_2/N_1) = ?$

Solution: we have Boltzmann factor, $\frac{N_2}{N_1} = e^{\frac{-\Delta E}{kT}}$ (note: use [shift + ln] for e in calculator)

$$\left(\frac{N_2}{N_1}\right) = e^{\frac{-hc}{\lambda kT}} = e^{\left(\frac{-6.63 \times 10^{-34} \times 3 \times 10^8}{694.3 \times 10^{-9} \times 1.38 \times 10^{-23} \times 300}\right)} = 8.87 \times 10^{-31}$$

$\therefore$ The ratio of population, $\left(\frac{N_2}{N_1}\right) = 8.87 \times 10^{-31}$

9. A laser beam at thermal equilibrium temperature 300K has two energy levels with a wavelength separation of $1\mu\text{m}$. find the ratio of population densities of the upper and lower levels.

(Jan 2018-17S)

(Refer problem no.8 same type)

$$\left(\text{Answer } \left(\frac{N_2}{N_1}\right) = 1.365 \times 10^{-21}\right)$$

10. A semiconductor has a peak emission radiation of wavelength of $1.24\mu\text{m}$. what is the band gap energy value in eV?

Data: wavelength of light emitted, $\lambda = 1.24\mu\text{m}$ or $\lambda = 1.24 \times 10^{-6}\text{m}$

To find: Band gap value $E_g = ?$

Solution: Band gap value $E_g = E_2 - E_1 = h\nu = \frac{hc}{\lambda}$

$$E_g = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.24 \times 10^{-6}}$$

$$E_g = 1.6 \times 10^{-19} \text{ J} = 1. \text{ eV}$$

$\therefore$ Band gap value E_g is 1.eV

OPTICAL FIBER:

11. The refractive indices of core and cladding are 1.50 and 1.48 respectively in an optical fiber.

Find the numerical aperture and angle of acceptance.

[Jan 2019]

Data: Refractive index of core, $n_1 = 1.50$

Refractive index of core, $n_2 = 1.48$

To find: Numerical aperture, $NA = ?$ and Angle of acceptance, $\theta = ?$

Solution: we have the relation for $NA = \sqrt{(n_1^2 - n_2^2)} = \sqrt{1.50^2 - 1.48^2} = 0.244$

$$\text{Angle of acceptance } \theta = \sin^{-1} \sqrt{(n_1^2 - n_2^2)} = \sin^{-1}(0.244) = 14.1^\circ$$

$\therefore$ The values of numerical aperture and angle of Acceptance angle are 0.244 and 14.1° respectively.

12. Calculate the refractive indices of core and cladding of a given optical fiber with numerical aperture of 0.22 and fractional index change variation 0.012.

(July 2022 -

21S)

Data: Numerical Aperture, $NA = 0.22$

Fractional index change, $\Delta = 0.012$

To find: Refractive index of core, $n_1 = ?$

Refractive index of core, $n_2 = ?$

Solution: we have the relation for $N.A = n_1 \sqrt{2\Delta}$

$$\text{or } n_1 = \frac{NA}{\sqrt{2\Delta}} = \frac{0.22}{\sqrt{2 \times 0.012}} = 1.42$$

we have the relation for $N.A = \sqrt{(n_1^2 - n_2^2)}$

$$\text{or } (NA)^2 = n_1^2 - n_2^2$$

$$\text{Or } n_2^2 = n_1^2 - (NA)^2$$

$$n_2^2 = 1.42^2 - (0.22)^2$$

$$n_2^2 = 1.496 \quad \text{or } n_2 = 1.40$$

$\therefore$ Refractive index of core, $n_1 = 1.42$ and Refractive index of core, $n_2 = 1.40$

13. The acceptance angle of an optical fiber is 30° when kept in an air find the angle of acceptance when it is in a medium of refractive index 1.33 (Dec 2017 | 17S)

Data: Refractive index of core, $n'_0 = 1.33$

Angle of acceptance, $\theta = 30^\circ$

To find: Angle of acceptance while the fiber is in the refracting medium, $\theta'_0 = 1.33$

Solution: we have the equation, $\sin \theta_0 = \frac{\sqrt{(n_1^2 - n_2^2)}}{n_0}$ (1)

When the surrounding medium is air, then $n_0 = 1$ and $\theta_0 = 30^\circ$

$$\sin 30^\circ = \frac{\sqrt{(n_1^2 - n_2^2)}}{1}$$

$$0.5 = \sqrt{(n_1^2 - n_2^2)} \dots \dots \dots (2)$$

When the surrounding medium is of the refractive index 1.33, let θ'_0 be the acceptance angle.

$$\text{Now eqn(1) becomes, } \sin \theta'_0 = \frac{\sqrt{(n_1^2 - n_2^2)}}{n'_0} = \frac{\sqrt{(n_1^2 - n_2^2)}}{1.33}$$

$$1.33 \sin \theta'_0 = \sqrt{(n_1^2 - n_2^2)} \dots \dots \dots (3)$$

$$\therefore \text{ from equation(2) and (3), } 1.33 \sin \theta'_0 = 0.5$$

$$\theta'_0 = \sin^{-1} \left(\frac{0.5}{1.33} \right) = 22^\circ$$

$\therefore$ The Acceptance angle when the fiber is in the medium of refractive index 1.33 is 22°

14. Calculate the V-number for a fiber of core of diameter $40\mu\text{m}$ and with refractive indices of 1.55 and 1.50 respectively for core and cladding. When the wavelength of the propagating wave is 1400nm . Also calculate the number of modes that the fiber can support for propagation. Assume that the fiber is in air. (July 2019 - 18S)

Data: Refractive index of core, $n_1 = 1.55$

Refractive index of core, $n_2 = 1.50$

Diameter of core, $d = 40\mu\text{m} = 40 \times 10^{-6}\text{m}$

Wavelength of the propagating light, $\lambda = 1400\text{nm} = 1400 \times 10^{-9}\text{m}$

To find: Numerical aperture, $NA=?$ and V-number, $V=?$

Solution: we have the relation for $NA = \sqrt{(n_1^2 - n_2^2)} = \sqrt{1.55^2 - 1.50^2} = 0.3905$

$$V - \text{number}, V = \frac{\pi d}{\lambda} \sqrt{(n_1^2 - n_2^2)}$$

$$V = \frac{\pi \times 40 \times 10^{-6}}{1400 \times 10^{-9}} \times 0.3905 = 35$$

$$\therefore \text{The number of modes of the fiber support}, N = \frac{V^2}{2} = \frac{35^2}{2} = 612$$

15. Estimate the attenuation in an optical fiber of length 500m when a light signal of power 100mW emerges out of fiber with a power 90 mW. (Jan 2020 | Aug 2021 | Feb 2022 – 18S)

Data: Length of the optical fiber = 500m = 0.5km

Input power of the signal, $P_{in} = 100\text{mW} = 100 \times 10^{-3}\text{W}$.

Output power of the signal, $P_{out} = 90\text{mW} = 90 \times 10^{-3}\text{W}$.

To find: Fiber attenuation, $\alpha=?$

Solution: The fiber attenuation α is given by, $\alpha = -\frac{10}{L} \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \text{ dB/km}$

$$\alpha = -\frac{10}{L} \log_{10} \left(\frac{90 \times 10^{-3}}{100 \times 10^{-3}} \right) = 0.915 \text{ dB/km}$$

$$\therefore \text{Fiber Attenuation} = 0.915 \text{ dB/km}$$

16. Calculate the number of modes that can propagate inside an optical fiber with the given data below. Refractive index of core is 1.53 and cladding is 1.50. Radius of core is 50 μm . Wavelength of the light propagating through the optical fiber is 1 μm . (Jan.2021 -18S)

Data: Refractive index of core, $n_1 = 1.53$

Refractive index of core, $n_2 = 1.50$

Diameter of core, $d = 2 \text{ times of radius, i.e } = 2r = 2 \times 50\mu\text{m} = 100\mu\text{m} = 100 \times 10^{-6}\text{m}$

Wavelength of the propagating light, $\lambda = 1\mu\text{m} = 1 \times 10^{-6}\text{m}$

To find: Numerical aperture, $NA=?$ And V-number, $V=?$

Solution: we have the relation for $NA = \sqrt{(n_1^2 - n_2^2)} = \sqrt{1.53^2 - 1.50^2} = 0.3014$

$$V - \text{number}, V = \frac{\pi \times d}{\lambda} \sqrt{(n_1^2 - n_2^2)}$$

$$V = \frac{\pi \times 100 \times 10^{-6}}{1 \times 10^{-6}} \times 0.3014 = 94.63$$

$$\therefore \text{The number of modes of the fiber support}, N = \frac{V^2}{2}$$

$$N = \frac{94.63^2}{2} = 4478.3$$

$$\therefore \text{The number of modes of the fiber support}, N = 4478.3$$

17. The attenuation of light in an optical fiber is 3.6 dB/km, what fraction of its initial intensity remains after: i) 1 km ii) 3km? [Dec 2014 | July.2022 -18S]

Data: *Fiber attenuation, $\alpha = 3.6$ dB/km*

To find: (i) *After 1 km distance, $\left(\frac{P_{out}}{P_{in}}\right) = ?$*
(ii) *After 3 km distance, $\left(\frac{P_{out}}{P_{in}}\right) = ?$*

Solution: we have, $\alpha = -\frac{10}{L} \log_{10} \left(\frac{P_{out}}{P_{in}}\right)$

Where, L the length of the fiber is expressed in km.

$$\therefore \log_{10} \left(\frac{P_{out}}{P_{in}}\right) = -\left(\frac{\alpha L}{10}\right)$$

$$\text{or } \left(\frac{P_{out}}{P_{in}}\right) = 10^{-\left(\frac{\alpha L}{10}\right)}$$

$$(i) \text{ At 1km distance, } L = 1 \quad \left(\frac{P_{out}}{P_{in}}\right) = 10^{-\left(\frac{3.6}{10}\right)} = 0.436$$

$$(i) \text{ At 3km distance, } L = 3 \quad \left(\frac{P_{out}}{P_{in}}\right) = 10^{-\left(\frac{3.6 \times 3}{10}\right)} = 0.0832$$

$$\therefore (i) \text{ After 1 km distance, } \left(\frac{P_{out}}{P_{in}}\right) = 0.436 \text{ and } (ii) \text{ After 3 km distance, } \left(\frac{P_{out}}{P_{in}}\right) = 0.0832$$

18. Calculate the NA, Relative RI, V number and the number of modes in an optical fiber of core diameter of 50 μm and the core and cladding R.I are 1.41 and 1.40 respectively. Given wavelength of source 820nm. (Feb 2022 - 18S)

Data: *Refractive index of core, $n_1 = 1.41$*

Refractive index of core, $n_2 = 1.40$

Diameter of core, $d = 50 \mu\text{m} = 50 \times 10^{-6} \text{m}$

Wavelength of the propagating light, $\lambda = 820 \text{nm} = 820 \times 10^{-9} \text{m}$

To find: Numerical aperture, $NA = ?$ and V-number, $V = ?$

Solution: we have the relation for $NA = \sqrt{(n_1^2 - n_2^2)} = \sqrt{1.41^2 - 1.40^2} = 0.1676$

$$V - \text{number, } V = \frac{\pi d}{\lambda} \sqrt{(n_1^2 - n_2^2)}$$

$$V = \frac{\pi \times 50 \times 10^{-6}}{820 \times 10^{-9}} \times 0.1676 = 32$$

$$\therefore \text{The number of modes of the fiber support, } N = \frac{V^2}{2} = \frac{32^2}{2} = 512$$

(Refer problem no.16 same type) (Answer $NA = 0.1676$, $\Delta = 7.093 \times 10^{-3}$ and $V = 512$)

MODULE 4 – ELECTRICAL CONDUCTIVITY OF METALS (FOR ECE AND CSE STREAM)

FERMI ENERGY:

1. Calculate the Fermi velocity and the mean free path for the conduction electrons in silver, given that its Fermi energy is 5.5 eV and the relaxation time for electrons is 3.83×10^{-14} s.

Data: *Fermi Energy, $E_F = 5.5\text{eV} = 5.5 \times 1.602 \times 10^{-19}\text{J}$* (Jan 2018 -17S)
Relaxation time for electrons, $\tau = 3.83 \times 10^{-14}$ s

Mass of electron, $m = 9.11 \times 10^{-31}\text{kg}$

To find: Fermi velocity, $v_f = ?$ and Mean free path, $\lambda = ?$

Solution: we have, $v_f = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2 \times 5.5 \times 1.602 \times 10^{-19}}{9.11 \times 10^{-31}}} = 1.396 \times 10^6 \text{ m/s}$
Mean free path, $\lambda = v_f \times \tau = 1.396 \times 10^6 \times 3.83 \times 10^{-14} = 5.518 \times 10^{-8} \text{ m}$

*Fermi velocity and mean free path for electrons in silver have the values
 $1.396 \times 10^6 \text{ m/s}$ and $5.518 \times 10^{-8} \text{ m}$ i.e $518\text{\AA}$.*

2. Find the temperature of which there is 1% probability that a state with an energy 0.5eV above the Fermi energy is occupied. (Jan 2019 -18S | July 2022 - 21S)

Data: *Probability, $f(E) = 1\% = 0.01$*

Energy above E_F i.e., $(E - E_F) = 0.5\text{eV} = 0.5 \times 1.602 \times 10^{-19}\text{J}$

To find : *Temperature at which the above data holds good, $T = ?$*

Solution: we have $f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$
 $\frac{E-E_F}{kT} = \frac{0.5 \times 1.602 \times 10^{-19}}{1.38 \times 10^{-23} \times T} = \frac{5797}{T}$

Substitute the above value, we have $0.01 = \frac{1}{e^{\frac{5797}{T}} + 1}$

$$e^{\frac{5797}{T}} = \frac{1}{0.01} - 1 = 99$$

Taking natural log on both sides we have, $\frac{5797}{T} = \ln 99$ ($\therefore \ln 99 = 4.595$)

$$\therefore T = \frac{5797}{4.595} = 1261.1\text{K}$$

$\therefore$ Temperature at which the above data holds good, $T = 1261.1\text{K}$

3. Calculate the probability of an electron occupying an energy level 0.02eV above the Fermi level at 200K and 400K in a material. (Jan 2020 | Jan 2021 -18S)

Data: $E - E_F = 0.02\text{eV} = 0.02 \times 1.602 \times 10^{-19}\text{J}$

To find: (i) $f(E)$ at 200K = ?

(ii) $f(E)$ at 400K = ?

Solution: (i) Evaluation of $f(E)$ at 200K = ?

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1} = \frac{1}{e^{\frac{0.02 \times 1.602 \times 10^{-19}}{1.38 \times 10^{-23} \times 200}} + 1} = \frac{1}{e^{1.1594} + 1} = \frac{1}{3.188 + 1} = \frac{1}{4.188} = 0.24$$

$\therefore f(E)$ at 200K = 0.24

(ii) Evaluation of $f(E)$ at $400K = ?$

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1} = \frac{1}{e^{\frac{0.02 \times 1.602 \times 10^{-19}}{1.38 \times 10^{-23} \times 400}} + 1} = \frac{1}{e^{0.5797} + 1} = \frac{1}{1.7855 + 1} = \frac{1}{2.7855} = 0.36$$

$$\therefore f(E) \text{ at } 400K = 0.36$$

$$\therefore \text{Evaluation of } f(E) \text{ at } 200K = 0.24$$

$$\therefore \text{Evaluation of } f(E) \text{ at } 400K = 0.36$$

4. Find the temperature of which there is 1% probability that a state with an energy 0.2eV above Fermi level is occupied. (Feb 2022 - 21S)

(Refer problem no.2 same type)

(Answer, $T = 505K$)

5. Calculate the probability of an electron occupying an energy level 0.02eV above the Fermi level and that in an energy level 0.02eV below the Fermi level at 200K. (Jan 2020 | Jan 2021 -18S)

Data: Energy level above the Fermi level, at 200K

$$E - E_F = 0.02\text{eV} = 0.02 \times 1.602 \times 10^{-19} \text{J}$$

To find:

$$(i) f(E) \text{ at } 200K = ?$$

Solution: (i) Evaluation of $f(E)$ at $200K = ?$ (Energy level above the Fermi level)

(ii) Evaluation of $f(E)$ at $200K = ?$ (Energy level below the Fermi level)

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1} = \frac{1}{e^{\frac{0.02 \times 1.602 \times 10^{-19}}{1.38 \times 10^{-23} \times 200}} + 1} = \frac{1}{e^{1.1594} + 1} = \frac{1}{3.188 + 1} = \frac{1}{4.188} = 0.24$$

$$\therefore f(E) \text{ at } 200K = 0.24$$

(i) Evaluation of $f(E)$ at $200K = ?$ (Energy level below the Fermi level)

Now, for the case of electron, occupying the energy level 0.02eV below the Fermi level E_F , at 200K,

We have $\Delta E = E_F - E = 0.02\text{eV}$

$$\therefore E - E_F = -0.02\text{eV} = -0.02 \times 1.602 \times 10^{-19} \text{J}$$

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1} = \frac{1}{e^{\frac{0.02 \times 1.602 \times 10^{-19}}{1.38 \times 10^{-23} \times 200}} + 1} = \frac{1}{e^{-1.1594} + 1} = \frac{1}{0.3137 + 1} = \frac{1}{1.3137} = 0.76$$

$$\therefore f(E) = 0.24 \text{ for } E = 0.02\text{eV above } E_F$$

$$\therefore f(E) = 0.76 \text{ for } E = 0.02\text{eV below } E_F$$

SEMICONDUCTORS AND HALL EFFECT: (FOR ECE STREAM ONLY)

6. The electron concentration in a semiconductor is $5 \times 10^{17} \text{m}^{-3}$. Calculate the conductivity of the material if the drift velocity of electron is 350ms^{-1} in an electric field of 1000Vm^{-1} .

(Jan 2018 -17S)

Data: Electron concentration in a semiconductor, $n_e = 5 \times 10^{17} \text{m}^{-3}$

Drift velocity of electron, $v_d = 350 \text{ms}^{-1}$

Electric field, $E = 1000 \text{Vm}^{-1}$

Charge of an electron, $e = 1.602 \times 10^{-19} \text{J}$

To find: *Conductivity of the material due to electron, $\sigma_e = ?$*

Solution: *Conductivity of the material due to electron, $\sigma_e = n_e e \mu_e \dots \dots \dots (1)$*

$$\text{Where, } \mu_e = \frac{v_d}{E} \quad \therefore \mu_e = \frac{v_d}{E} = \frac{350}{1000} = 0.35 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$$

Substitute the value in eqn (1) $\sigma_e = 5 \times 10^{17} \times 1.602 \times 10^{-19} \times 0.35 = 0.028 \text{ Ohm} - \text{meter}$

$\therefore$ Conductivity of the material due to electron, $\sigma_e = 0.028 \text{ Ohm} - \text{meter}$

7. The Resistivity of intrinsic germanium at 27°C is equal to 0.47 ohm-m. Assuming the electron and hole mobilities as 0.38 and 0.18 m²/V-Sec respectively. Calculate the intrinsic carrier density.

(Jan 2019 -18S | Feb 2022 -21S)

Data: *Resistivity of intrinsic germanium, $\rho_i = 0.47 \text{ ohm} - \text{m}$*

Electron mobilities, $\mu_e = 0.38 \text{ m}^2/\text{V} - \text{Sec}$

Hole mobilities, $\mu_h = 0.18 \text{ m}^2/\text{V} - \text{Sec}$

Charge of an electron, $e = 1.602 \times 10^{-19} \text{ J}$

To find: *Carrier density, $n_i = ?$*

Solution: *Carrier density, $n_i = \frac{1}{\rho_i e (\mu_e + \mu_h)} = \frac{1}{0.47 \times 1.602 \times 10^{-19} (0.38 + 0.18)} = 2.3716 \times 10^{19} / \text{m}^3$*

$\therefore$ Carrier density, $n_i = 2.3716 \times 10^{19} / \text{m}^3$

8. The intrinsic charge carrier concentration of germanium is $2.4 \times 10^{19} / \text{m}^3$, calculate its resistivity if mobility of electrons and holes respectively are 0.39 m²/vs and 0.19 m²/vs.

Data: *Intrinsic Carrier concentration, $n_i = 2.4 \times 10^{19} / \text{m}^3$ (July 2019 | Aug 2020 - 18S)*

Electron mobilities, $\mu_e = 0.39 \text{ m}^2/\text{V} - \text{Sec}$

Hole mobilities, $\mu_h = 0.19 \text{ m}^2/\text{V} - \text{Sec}$

Charge of an electron, $e = 1.602 \times 10^{-19} \text{ J}$

To find: *Resistivity of intrinsic germanium sample, $\rho_i = ?$*

Solution: *we have the relation for conductivity σ_i as, $\sigma_i = n_i e (\mu_e + \mu_h)$*

$$\begin{aligned} \text{But Resistivity, } \rho_i &= \frac{1}{\sigma_i} = \frac{1}{n_i e (\mu_e + \mu_h)} \\ &= \frac{1}{2.4 \times 10^{19} \times 1.602 \times 10^{-19} (0.39 + 0.19)} \\ &= 0.449 \text{ Ohm} - \text{meter} \end{aligned}$$

$\therefore$ Resistivity of intrinsic germanium sample, $\rho_i = 0.449 \text{ Ohm} - \text{meter}$

9. Calculate the concentration at which donor atoms need to be added to a silicon semiconductor. So that it results in n-type semi conductivity of $2.2 \times 10^{-4} \text{ mho/m}$ and the mobility of electron being $1.25 \times 10^{-3} \text{ m}^2/\text{VS}$. (Aug 2020 -18S)

Data: *conductivity, $\sigma_e = 2.2 \times 10^{-4} \text{ mho/m}$*

Electron mobilities, $\mu_e = 1.25 \times 10^{-3} \text{ m}^2/\text{V} - \text{Sec}$

Charge of an electron, $e = 1.602 \times 10^{-19} \text{ J}$

To find: *Concentration of donor impurity, $N_e = ?$*

Solution: *The conductivity, $\sigma_e = N_e e \mu_e$*

Or rearranging this we get, $N_e = \frac{\sigma_e}{e \mu_e} = \frac{2.2 \times 10^{-4}}{1.602 \times 10^{-19} \times 1.25 \times 10^{-3}} = 1.0986 \times 10^{18} / \text{m}^3$

$\therefore$ The Concentration of donor impurity, $N_e = 1.0986 \times 10^{18} / \text{m}^3$

10. The Hall coefficient of a material is $-3.68 \times 10^{-5} \text{ m}^3/\text{C}$. Identify the type of charge carriers and calculate the carrier Concentration. (Feb -2022 -18S)

Data: The Hall coefficient of a material is $-3.68 \times 10^{-5} \text{ m}^3/\text{C}$

To find: Carrier concentration, $n = ?$

Solution: *since the sign of R_H is negative as per data, the charge carriers are electrons.*

We have, $R_H = \frac{1}{\rho} = \frac{1}{n \times e}$

Or rearranging $n = \frac{1}{R_H \times e} = \frac{1}{3.68 \times 10^{-5} \times 1.602 \times 10^{-19}} = 1.7 \times 10^{23} / \text{m}^3$.

$\therefore$ Carrier concentration, $n = 1.7 \times 10^{23} / \text{m}^3$

11. The conductivity and Hall coefficient of an n-type semiconductor are $112 / \Omega \text{m}$ and $1.25 \times 10^{-3} \text{ m}^3/\text{C}$ respectively. Calculate the charge carrier concentration and electron mobility. (July 2022 -18S)

Data: Conductivity of an n - type semiconductor, $\sigma_e = 112 / \Omega \text{m}$

Charge of an electron, $e = 1.602 \times 10^{-19} \text{ J}$

Hall coefficient, $R_H = 1.25 \times 10^{-3} \text{ m}^3/\text{C}$

To find: Carrier concentration, $n_e = ?$

Electron mobility, $\mu_e = ?$

Solution: w. k. t Hall coefficient, $R_H = \frac{1}{n_e \times e}$

Rearranging this, $n_e = \frac{1}{R_H \times e} = \frac{1}{1.25 \times 10^{-3} \times 1.602 \times 10^{-19}} = 5 \times 10^{21} / \text{m}^3$

W.K.T Electrical conductivity due to electrons, $\sigma_e = n_e \times \mu_e \times e$

Electron mobility, $\mu_e = \frac{\sigma_e}{n_e \times e} = \frac{112}{5 \times 10^{21} \times 1.602 \times 10^{-19}} = 0.14 \text{ m}^2 \text{v}^{-1} \text{s}^{-1}$

$\therefore$ Carrier concentration, $n_e = 5 \times 10^{21} / \text{m}^3$ and Electron mobility, $\mu_e = 0.14 \text{ m}^2 \text{v}^{-1} \text{s}^{-1}$

DIELECTRICS: (ONLY FOR ECE STREAM)

12. The Dielectric constant of sulphur is 3.4. Assuming a cubic lattice for its structure, calculate the electronic polarizability of sulphur. (Given, density of sulphur, 2.07 gm/cc and atomic weight 32.07). (July 2019 -18S)

Data: Dielectric constant, $\epsilon_r = 3.4$

Density of sulphur, $D = 2.07 \text{ gm/cc} = 2.07 \times 10^3 \text{ kg/m}^3$.

Atomic Weight = 32.07

Crystal structure of Sulphur is Cubic.

To find : Electronic polarisability of sulphur, $\alpha_e = ?$

Solution: w. k. t Clausius Mossotti equation, $\left(\frac{\epsilon_r - 1}{\epsilon_r + 2} \right) = \frac{N \alpha_e}{3 \epsilon_0}$

Or $\alpha_e = \frac{3 \epsilon_0}{N} \left[\frac{\epsilon_r - 1}{\epsilon_r + 2} \right] \dots \dots \dots (1)$

Now, N is the no of atoms/unit can be written as

$$N = \frac{N_A \times D}{\text{atomic weight}} = \frac{6.025 \times 10^{26} \times 2.07 \times 10^3}{32.07} = 3.89 \times 10^{28} / \text{m}^3$$

By substituting the values of N in eqn (1) $\alpha_e = \frac{3 \times 8.856 \times 10^{-12}}{3.89 \times 10^{28}} \left[\frac{3.4 - 1}{3.4 + 2} \right] = 3.035 \times 10^{-40} \text{ Fm}^2$

$\therefore$ Electronic polarisability of sulphur, $\alpha_e = 3.035 \times 10^{-40} \text{ Fm}^2$

13. An elemental solid dielectric material has polarizability $7 \times 10^{40} \text{ Fm}^2$. Assuming the internal field to be Lorentz field, calculate the dielectric constant for the material if the material has $3 \times 10^{28} \text{ atoms/m}^3$

(Jan 2020 | July 2022 -18S)

Data: polarizability, $\alpha_e = 7 \times 10^{40} \text{ Fm}^2$
 No of $\frac{\text{atoms}}{\text{m}^3}$, $N = 3 \times 10^{28} \frac{\text{atoms}}{\text{m}^3}$
 The internal field is Lorentz field.

To find: Dielectric constant of the material $\epsilon_r = ?$

Solution: w. k. t Clausius Mossotti equation, $\left(\frac{\epsilon_r - 1}{\epsilon_r + 2}\right) = \frac{N\alpha_e}{3\epsilon_0}$

$$\left(\frac{\epsilon_r - 1}{\epsilon_r + 2}\right) = \frac{3 \times 10^{28} \times 7 \times 10^{40}}{3 \times 8.856 \times 10^{-12}} = 0.7906$$

$$(\epsilon_r - 1) = (\epsilon_r + 2) \times 0.7906$$

$$(\epsilon_r - 1) = 0.7906\epsilon_r + 1.5812$$

$$\epsilon_r - 0.7906\epsilon_r = 2.5812$$

$$\epsilon_r(1 - 0.7906) = 2.5812$$

$$\epsilon_r = \frac{2.5812}{0.2094} = 12.33$$

$\therefore$ Dielectric constant of the material $\epsilon_r = 12.33$

14. Find the polarization produced in a dielectric medium of relative permittivity 15 in presence of an electric field of strength 500V/m.

(Jan 2021 -18S)

Data: Applied electric field, $E = 500 \text{ V/m}$,
 Relative permittivity, $\epsilon_r = 15$
 Dielectric constant, $\epsilon_0 = 8.856 \times 10^{-12} \text{ farad/m}$

To find: Polarisation, $P = ?$

Solution: we have $P = \epsilon_0(\epsilon_r - 1)E$

$$P = 8.856 \times 10^{-12} (15 - 1) 500 = 6.198 \times 10^{-8} \text{ cm}^{-2}$$

$\therefore$ Polarisation P , produced in a dielectric medium, $\epsilon_r = 6.198 \times 10^{-8} \text{ cm}^{-2}$

15. If a NaCl crystal is subjected to an electric field of 1000 V/m and the resulting Polarization is $4.3 \times 10^{-8} \text{ C/m}^2$. Calculate the dielectric constant of NaCl.

(Aug 2021 | Feb -2022 -18S)

Data: Applied electric field, $E = 1000 \text{ V/m}$,
 Resulting polarisation, $P = 4.3 \times 10^{-8} \text{ C/m}^2$

Dielectric constant, $\epsilon_0 = 8.856 \times 10^{-12} \text{ farad/m}$

To find: The dielectric constant of NaCl, $\epsilon_r = ?$

Solution: we have $P = \epsilon_0(\epsilon_r - 1)E$

$$\epsilon_r = \frac{P}{\epsilon_0 \times E} + 1 = \frac{4.3 \times 10^{-8}}{8.856 \times 10^{-12} \times 1000} + 1 = 5.855$$

$\therefore$ The Dielectric constant of NaCl, $\epsilon_r = 5.855$

16. The dielectric constant of He gas at NTP is 1.0000684. Calculate the electronic polarisability of atoms if the gas contains $2.7 \times 10^{28} \text{ atoms/m}^3$.

(July 2022 - 21S)

Data: Dielectric constant of He gas at NTP, $\epsilon_r = 1.0000684$
 Density of He atoms, $N = 2.7 \times 10^{28} \text{ atoms/m}^3$

To find: Electronic polarisability, $\alpha_e = ?$

Solution: The Electronic polarisability, $\alpha_e = 4\pi\epsilon_0 R^3$

$$\text{Or } R = \left(\frac{\alpha_e}{4\pi\epsilon_0} \right)^{\frac{1}{3}} \dots \dots \dots (1)$$

$$\alpha_e = \frac{\epsilon_0(\epsilon_r - 1)}{N} = \frac{8.85 \times 10^{-12}(1.0000684 - 1)}{2.7 \times 10^{28}} = 2.24210^{-44} \text{ Fm}^2$$

$$\text{Using equation(1), we have } R = \left(\frac{\alpha_e}{4\pi\epsilon_0} \right)^{\frac{1}{3}} = \left(\frac{2.24210^{-44}}{4\pi \times 8.85 \times 10^{-12}} \right)^{\frac{1}{3}} = 5.8635 \times 10^{-12} \text{ m}$$

∴ The electronic polarisability of He atoms at NTP is equal to $2.24210^{-44} \text{ Fm}^2$

and the Radius of Helium atom is $5.8635 \times 10^{-12} \text{ m}$

MODULE 5 – MATERIALS CHARACTERIZATION TECH AND INSTRUMENTATION (ONLY FOR CIVIL STREAM)

BRAGG'S LAW:

1. First order spectrum is formed when X-rays of wavelength $1.5\text{\AA}$ is incident on crystal at 12° . Calculate the interplanar spacing of the crystal.

Data: X – rays of wavelength, $\lambda = 1.5\text{\AA} = 1.5 \times 10^{-10}\text{m}$

Order of diffraction, $n = 1$

Glancing angle, $\theta = 12^\circ$

To find: The Interplanar spacing of the crystal, $d = ?$

Solution: $2d\sin\theta = n\lambda$

Or rearranging, $d = \frac{n\lambda}{2\sin\theta}$

$$d = \frac{1 \times 1.5 \times 10^{-10}}{2 \times \sin 12^\circ} = 3.6 \times 10^{-10}\text{m}$$

$\therefore$ The Interplanar spacing of the crystal, $d = 3.6 \times 10^{-10}\text{m}$ or $3.6\text{\AA}$

2. Using a Bragg's spectrometer, the glancing angle for the first order spectrum was observed to be equal to 6° . Find the wavelength of the X-rays if $d = 2.82\text{\AA}$.

Data: The Interplanar spacing of the crystal, $d = 2.82\text{\AA} = 2.82 \times 10^{-10}\text{m}$

Order of diffraction, $n = 1$

Glancing angle, $\theta = 6^\circ$

To find: Wavelength of X – rays, $\lambda = ?$

Solution: From Bragg's law, we have $2d\sin\theta = n\lambda$

$$\text{or } \lambda = \frac{2d\sin\theta}{n} = \frac{2 \times 2.82 \times 10^{-10} \times \sin 6^\circ}{1} = 0.5895 \times 10^{-10}\text{m}$$

$\therefore$ Wavelength of X – rays, $\lambda = 0.5895 \times 10^{-10}\text{m}$ or $0.5895\text{\AA}$

3. Calculate the glancing angle for incidence of X-rays of wavelength $0.58\text{\AA}$ on the plane (132) of NaCl which results in second order diffraction maxima taking the lattice constant as $3.81\text{\AA}$.

Data: X – rays of wavelength, $\lambda = 0.58\text{\AA} = 0.58 \times 10^{-10}\text{m}$

(Jan 2018 -17S)

$(h,k,l) = (1, 3, 2)$

Order of diffraction, $n = 2$

Lattice constant $a = 3.81\text{\AA} = 3.81 \times 10^{-10}\text{m}$

To find: Glancing angle, $\theta = ?$

Solution: $2d\sin\theta = n\lambda$ (1)

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{3.81 \times 10^{-10}}{\sqrt{1^2 + 3^2 + 2^2}} = 1.0182 \times 10^{-10}\text{m}$$

$$\text{Rearranging eqn (1), } \theta = \sin^{-1} \left(\frac{n\lambda}{2d} \right) = \sin^{-1} \left(\frac{2 \times 0.58 \times 10^{-10}}{2 \times 1.0182 \times 10^{-10}} \right) = 34.72^\circ$$

$\therefore$ **Glancing angle, $\theta = 34.72^\circ$**

4. First order Bragg reflection occurs when a monochromatic beam of X-rays of wavelength $0.675\text{\AA}$ is incident on a crystal at a glancing angle of $4^\circ 51'$. What is the glancing angle for third order Bragg reflection to occur?

Data: X-rays of wavelength, $\lambda = 0.675\text{\AA} = 0.675 \times 10^{-10}\text{m}$

Order of diffraction, $n = 1$

Glancing angle, $\theta_1 = 4^\circ 51' = 4.85^\circ$

To find: For Third order reflection ($n = 3$), $\theta_3 = ?$

The Interplanar spacing of the crystal, $d = ?$

Solution: For first order $2d \sin \theta_1 = n\lambda$

Or rearranging, $d = \frac{n\lambda}{2 \sin \theta_1}$

$$d = \frac{1 \times 0.675 \times 10^{-10}}{2 \times \sin 4.85^\circ} = 4 \times 10^{-10}\text{m}$$

$\therefore$ The Interplanar spacing of the crystal, $d = 4 \times 10^{-10}\text{m}$ or $4\text{\AA}$

$\therefore$ For third order reflection, $2d \sin \theta_3 = 3\lambda$

$$\text{or } \theta_3 = \sin^{-1} \left(\frac{n\lambda}{2d} \right) = \sin^{-1} \left(\frac{3 \times 0.675 \times 10^{-10}}{2 \times 4 \times 10^{-10}} \right) = 14.6625 = 14^\circ 39' 45''$$

$\therefore$ **The Third order reflection occurs at a glancing angle of $14^\circ 39' 45''$**

5. GaAs has its principal planes separated at $5.6534\text{\AA}$. The first order Bragg's reflection is located at $13^\circ 40'$. Calculate the wavelength of the X-ray and angle for second order Bragg reflection.

Data: Interplanar distance $d = 5.6534\text{\AA} = 5.6534 \times 10^{-10}\text{m}$

(July-2022 -21S)

Order $n = 1$

Angle of reflection $\theta = 13^\circ 40'$

To find: wavelength of the X-rays, $\lambda = ?$ and

Angle for second order Bragg reflection, $\theta_2 = ?$

Solution: For First order $n = 1$

$$2d \sin \theta = n\lambda$$

$$\lambda = \frac{2d \sin \theta}{n} = \frac{2 \times 5.6534 \times 10^{-10} \times \sin 13^\circ 40'}{1} = 2.6715 \times 10^{-10}\text{m}$$

For second order, $n = 2$ $2d \sin \theta = n\lambda$

$$\therefore \theta = \sin^{-1} \left(\frac{n\lambda}{2d} \right)$$

$$\theta = \sin^{-1} \left(\frac{2 \times 2.6715 \times 10^{-10}}{2 \times 5.6534 \times 10^{-10}} \right) = 28^\circ 12'$$

$\therefore$ Angle for second order Bragg reflection, $\theta = 28^\circ 12'$

SCHERER'S EQUATION:

6. Determine the crystal size when the peak width is 0.5° and peak position 30° for a cubic crystal. The wavelength of X-rays used is $100\text{\AA}$ and Scherer's constant $K=0.92$. (Feb 2022 -21S)

Data: Wavelength of the X-rays, $\lambda = 100\text{\AA} = 100 \times 10^{-10}\text{m}$
 Peak width, $\beta = 0.5^\circ = 0.5 \times \frac{\pi}{180}$ (peak width need to convert in terms of radians)
 Peak position, $2\theta = 30^\circ$, or $\theta = 15^\circ$ or $\theta = 15 \times \frac{\pi}{180}$ (need to convert into radians)
 Scherer's constant $K = 0.92$

To find: Crystal size determination, $D = ?$

Solution: Crystal size determination, $D = \frac{K \times \lambda}{\beta \times \cos \theta}$

$$D = \frac{0.92 \times 100 \times 10^{-10}}{0.5 \times \frac{\pi}{180} \times \cos \left(15 \times \frac{\pi}{180} \right)} = 1.0542 \times 10^{-6}\text{m}$$

$\therefore$ Crystal size, $D = 1.0542 \times 10^{-6}\text{m}$ or $1.0542\mu\text{m}$

7. Determine the crystal size given the wavelength of X-rays 12nm , the peak width 0.5° and peak position 23° for a cubic crystal. Given $K=0.94$ (July 2022 -18S)

Data: wavelength of the X-rays, $\lambda = 12\text{nm} = 12 \times 10^{-9}\text{m}$
 Peak width, $\beta = 0.5^\circ = 0.5 \times \frac{\pi}{180}$ (peak width need to convert in terms of radians)
 peak position, $2\theta = 23^\circ$ then
 $\therefore \theta = 11.5^\circ$ or $\theta = 11.5 \times \frac{\pi}{180}$ (need to convert into radians)
 Scherer's constant $K = 0.94$

To find: Crystal size determination, $D = ?$

Solution: Crystal size determination, $D = \frac{K \times \lambda}{\beta \times \cos \theta}$

$$D = \frac{0.94 \times 12 \times 10^{-9}}{0.5 \times \frac{\pi}{180} \times \cos \left(11.5 \times \frac{\pi}{180} \right)} = 1.265 \times 10^{-6}\text{m}$$

$\therefore$ Crystal size, $D = 1.265 \times 10^{-6}\text{m}$ or $1.265\mu\text{m}$

8. Determine the wavelength of X-rays for the crystal size of $1.188 \times 10^{-6}m$, peak width is 0.5° and the peak position of 30° for a cubic crystal. Given that Scherer's constant $K=0.92$.

Data: Crystal size, $D = 1.188 \times 10^{-6}m$

Peak width, $\beta = 0.5^\circ = 0.5 \times \frac{\pi}{180}$ (peak width need to convert in terms of radians)

peak position, $2\theta = 30^\circ$ then

$\therefore \theta = 15^\circ$ or $\theta = 15 \times \frac{\pi}{180}$ (need to convert into radians)

Scherer's constant $K = 0.92$

To find: wavelength of the X – rays, $\lambda = ?$

Solution: $D = \frac{K \times \lambda}{\beta \times \cos \theta} \quad \therefore \text{Rearranging the above expression, } \lambda = \frac{D \times \beta \times \cos \theta}{K}$

$$\lambda = \frac{1.188 \times 10^{-6} \times \left(0.5 \times \frac{\pi}{180}\right) \times \cos\left(15 \times \frac{\pi}{180}\right)}{0.92} = 1.1268 \times 10^{-8}m$$

$\therefore$ Wavelength of the X – rays, $\lambda = 1.1268 \times 10^{-8}m$